

Master QA Final Testing, Bug-Fixing & Release Validation Report

Date of Audit: October 3, 2026 Auditor: Senior System QA & Release Auditor Project Baseline: Healthy Bite Custom PHP MVC	Target Database: MySQL 10.4 (healthy_bite) PHP Environment: PHP 8.5.10 (CLI / Server) Architecture: Custom MVC / PDO / Vanilla JS	FINAL VERDICT PASS (100%) Ready for Release
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1. Executive Test Execution Summary

Total Tests Executed	Passed	Failed	Defects Resolved	Regression Tests	Final Release Status
74	74 (100%)	0 (0%)	4 Fixed	All Suites Pass	PASS — APPROVED

2. Complete Register of All 74 Executed Tests

Every test was executed against the running PHP server, live MySQL database, and application services.

ID	Category	Test Name / Scope	Expected Behavior / Verification	Status
T-01	Environment	PHP 8.2+ Compatibility	PHP version is >= 8.2 (Running: PHP 8.5.10)	PASS
T-02	Environment	PDO MySQL Driver	pdo_mysql extension active and loaded	PASS
T-03	Environment	Live Database Connection	PDO connection to healthy_bite responds SELECT 1 = 1	PASS
T-04	Environment	.env Configuration	.env present, loaded and parseable via App\Core\Env	PASS
T-05	Environment	Core Constants	ROLE_*, ORDER_*, TABLE_*, FOOD_* constants defined	PASS
T-06	Static Analysis	PHP Syntax Validation	All 94 project PHP files pass php -l lint with zero errors	PASS
T-07	Static Analysis	JavaScript VM Compilation	All 14 public JS files pass Node.js VM syntax compilation	PASS
T-08	Database	17 Physical Tables	All 17 documented tables verified in information_schema	PASS
T-09	Database	208 Database Columns	Exact column count across all 17 tables matches 208	PASS
T-10	Database	26 Foreign Keys	All 26 foreign key constraints verified active	PASS
T-11	Database	Order-Payment 1:1 Mapping	payments.order_id UNIQUE key strictly enforced (Non_unique=0)	PASS
T-12	Database	Orphan Record Scan	Zero orphan rows across all 26 FK relations (0 orphans)	PASS
T-13	Unit: Pricing	Base + Variant + Custom Price	Base (100) + Var (20) + Custom (25) = Rs 145.00	PASS
T-14	Unit: Pricing	Line Total Scaling	Rs 145.00 x 3 = Rs 435.00 exact line total	PASS
T-15	Unit: Pricing	Order Totals & 5% GST	Subtotal Rs 435 + Tax Rs 21.75 = Rs 456.75	PASS
T-16	Unit: Nutrition	8 Macros & Caffeine Sum	All 8 macros summed: Cal, Prot, Carb, Fat, Fib, Sug, Sod, Caff	PASS
T-17	Unit: Nutrition	Caffeine in mg	Caffeine calculated in mg and never coerced to grams	PASS
T-18	Unit: Nutrition	Quantity Nutrition Scaling	Nutrition doubles accurately when quantity = 2	PASS
T-19	Unit: Nutrition	Strict NULL Preservation	Unmeasured fields retain NULL, not coerced to 0	PASS
T-20	Unit: QR	Valid QR Token Resolution	Active token resolves restaurant, branch, and table IDs	PASS

T-21	Unit: QR	Invalid QR Token Rejection	Random/expired token returns null context	PASS
T-22	Unit: Cart	Empty Cart Rejection	Empty cart payload throws InvalidArgumentException	PASS
T-23	Unit: Cart	Cross-Restaurant Isolation	Food from Restaurant A rejected under Restaurant B context	PASS
T-24	API	GET /api/health	Returns HTTP 200 with status=ok and database=connected	PASS
T-25	API	GET /api/restaurant	Returns HTTP 200 with active restaurant profile	PASS
T-26	API	GET /api/categories	Returns HTTP 200 with list of active menu categories	PASS
T-27	API	GET /api/foods	Returns HTTP 200 with complete menu items catalog	PASS
T-28	API	GET /api/foods/1	Returns HTTP 200 with Paneer Bowl details & variants	PASS
T-29	API	GET /api/foods/99999	Returns HTTP 404 for non-existent food item ID	PASS
T-30	API	POST /api/cart/validate	Returns HTTP 200 for valid items & required groups	PASS
T-31	API	Cart Server Recalculation	Server ignores client fake prices; computes from DB	PASS
T-32	Auth & RBAC	Owner Dashboard Protection	Unauthenticated /owner/dashboard redirects to /owner/login	PASS
T-33	Auth & RBAC	Admin Dashboard Protection	Unauthenticated /admin/dashboard redirects to /admin/login	PASS
T-34	Auth & RBAC	Invalid Credentials Check	Bad email/password fails authentication gracefully	PASS
T-35	Auth & RBAC	Removed Admin Orders Route	/admin/orders returns HTTP 404 (strictly eliminated)	PASS
T-36	Customer E2E	Order Creation Pipeline	Order placed with valid customer, table, and branch	PASS
T-37	Customer E2E	Order Item Snapshot Freeze	food_name, base_price, unit_price saved in snapshot	PASS
T-38	Customer E2E	Customization Snapshot Freeze	customization_name and price_adjustment saved in snapshot	PASS
T-39	Customer E2E	Payment Simulation (UPI)	Simulates payment, generates TXN ref, marks completed	PASS
T-40	Customer E2E	Status Advance on Payment	Order status advances from placed to accepted upon payment	PASS
T-41	Customer E2E	Order Tracking Polling	Live tracking finds order by order_number with status	PASS
T-42	Owner & Kitchen	Kitchen Kanban State Machine	Status advances: placed->accepted->preparing->ready->completed	PASS
T-43	Owner & Kitchen	Dining Tables 1-12 Active	All 12 dining tables present and queryable for Branch 1	PASS
T-44	Owner & Kitchen	Table Status Toggle	Table toggles available <-> occupied seamlessly	PASS
T-45	Owner & Kitchen	Review Reply Persistence	Owner reply persisted to reviews table with replied_at	PASS
T-46	Admin E2E	Super Admin Role Verification	Super Admin account verified with role_id = 1	PASS
T-47	Admin E2E	Restaurant Owner Role Verification	Owner account verified with role_id = 2	PASS
T-48	Admin E2E	Kitchen Staff Role Verification	Kitchen Staff account verified with role_id = 4	PASS
T-49	Admin E2E	Approved Restaurants Query	Platform lists approved active tenant restaurants	PASS
T-50	Security	SQL Injection: Search Filter	Malicious SQL in search handled via PDO prepared statements	PASS
T-51	Security	SQL Injection: Query Params	IDs and filters parameterized; zero SQL injection vectors	PASS
T-52	Security	XSS Output Escaping	HTML/JS tags escaped via htmlspecialchars in views	PASS
T-53	Security	CSRF Protection	State-changing POST actions protected by CSRF tokens	PASS
T-54	Security	IDOR Tenant Protection	Cross-restaurant cart and order modification blocked	PASS
T-55	Immutability	Catalog Price Independence	Updating food base_price does not alter historical order snapshot	PASS
T-56	Concurrency	Duplicate Payment Guard	Repeated payment request on paid order blocked gracefully	PASS
T-57	Dynamic Flow	Find Table with Active Token	BranchRepository::findTableByNumber locates table + token	PASS
T-58	Dynamic Flow	Find First Available Table	BranchRepository::getFirstAvailableTable returns active table	PASS
T-59	Dynamic Flow	Table Number Redirect	HTTP GET /table/12 redirects (302) to /menu?token=...	PASS

T-60	Dynamic Flow	Invalid Token Error Banner	HTTP GET /menu with invalid token displays error notice	PASS
T-61	Dynamic Flow	Category Auto-Slug Generation	CategoryRepository::createCategory generates slug automatically	PASS
T-62	Dynamic Flow	Category Deletion Guard	Category deletion blocked if active food items are attached	PASS
T-63	Dynamic Flow	Staff Record Update	StaffRepository::updateStaff dynamically updates member details	PASS
T-64	Dynamic Flow	Macro Sugar Aggregations	OrderRepository::getMacroStats computes total & avg sugar	PASS
T-65	Dynamic Flow	Checkout Table Dropdown	Checkout view renders verified active branch tables	PASS
T-66	Dynamic Flow	Owner Analytics Data Load	Analytics controller loads real AOV, revenue, orders	PASS
T-67	Dynamic Flow	Sugar Snapshot Integrity	OrderService creates order with accurate sugar snapshot	PASS
T-68	Catalog & Media	45 Unique Food Dish URLs	All 45 Unsplash dish image URLs return HTTP 200 OK	PASS
T-69	Catalog & Media	Distinct Photo Identifiers	All 45 food items use 100% distinct photo IDs	PASS
T-70	Catalog & Media	Food Dietary Types Valid	food_type enum values strictly vegetarian, vegan, etc.	PASS
T-71	Release	HTTP HEAD Method Route Matching	Router maps HEAD to GET; curl -I returns HTTP 200 OK	PASS
T-72	Release	Order Details Modal Retrieval	Order modal API retrieves items and customizations	PASS
T-73	Release	Menu Item Create/Edit/Delete	Full food item CRUD lifecycle tested and restored	PASS
T-74	Release	Legacy Features Clean Exclusion	No Call Servant or Platform Orders code exists	PASS

3. Defect Diagnosis and Resolution Register

During rigorous testing, four defects were identified, diagnosed to their root cause, corrected in source code, and verified through regression testing.

Bug ID	Severity	Component	Root Cause & Source File	Fix Applied & Retest Verification
BUG-01	MEDIUM	Router.php	HEAD requests skipped GET routes and returned 404. <i>File: app/Core/Router.php</i>	Mapped effective method: (\$requestMethod === 'HEAD') ? 'GET' : \$requestMethod. Retested: HTTP 200 OK on HEAD.
BUG-02	HIGH	PaymentService.php	Repeated payment on already paid order caused unhandled MySQL 1062 duplicate key error. <i>File: app/Services/PaymentService.php</i>	Added guards checking order_status === 'cancelled' and payment_status === 'completed'. Retested: HTTP 422 JSON returned cleanly.
BUG-03	HIGH	Database / Schema	59 historical order records referenced deleted customization IDs from past seeder truncation. <i>Database: order_item_customizations</i>	Restored 28 legacy customization definitions with is_available = 0 to satisfy FK. Retested: 0 orphans across all 26 FKs .
BUG-04	LOW	Dynamic Test Harness	Test cart payload omitted required customization groups, triggering valid cart exception. <i>File: tests/test_full_system_dynamic_flow.php</i>	Updated test harness to select 1 option per required group dynamically. Retested: 11/11 tests passed .

4. Defensive Security Testing Results

Attack Vector / Scope	Security Test Applied	Protection Mechanism	Result
SQL Injection (SQLi)	Injected payload "' OR '1'='1' -- " into search, query parameters, and IDs.	PDO prepared statements with parameterized placeholders across all repositories.	PASS
Cross-Site Scripting (XSS)	Injected "alert('XSS')" into names, notes, and customer review comments.	htmlspecialchars output escaping with ENT_QUOTES and UTF-8 encoding in views.	PASS
Cross-Site Request Forgery	Dispatched state-changing POST requests without valid CSRF session tokens.	Csrf::validate() token verification rejects unauthorized state changes.	PASS
IDOR & Multi-Tenant Isolation	Attempted to order Food from Restaurant 1 under Restaurant 2 context via API payload.	CartService and SQL queries enforce strict restaurant_id multi-tenant scoping.	PASS
Session & Auth Security	Attempted direct protected URL access to owner/admin portals without active session.	RestaurantMiddleware and AdminMiddleware redirect unauthenticated sessions to login.	PASS

5. Verified Database Schema Architecture

Table Name	Rows	Cols	FKs	Primary Key	Role / Constraints
admin	4	4	0	id	Legacy static administration table
branches	6	10	1	id	Restaurant dining locations
categories	16	10	1	id	Menu category hierarchy
customers	37	6	0	id	Registered and guest patrons
food_customizations	564	21	1	id	Dishes modifiers & adjustments
food_items	47	25	2	id	Main culinary catalog & macros
food_variants	94	18	1	id	Portion sizes & variant adjustments
order_item_customizations	66	15	2	id	Frozen customization snapshot
order_items	43	19	2	id	Frozen item price/macro snapshot
orders	34	16	4	id	Core order transactions & statuses
payments	21	8	1	id	Tender settlement (UNIQUE: order_id)
qr_tokens	12	8	3	id	Table dining session tokens
restaurant_tables	12	7	2	id	Physical dining tables (1-12)
restaurants	3	15	1	id	Tenant restaurant accounts
reviews	7	11	3	id	Customer ratings & owner replies
roles	4	6	0	id	RBAC roles (admin, owner, manager, staff)
users	8	9	2	id	Authenticated employee & admin logins

QA AUDITOR SIGN-OFF & CERTIFICATION This certifies that the Healthy Bite application has undergone comprehensive static analysis, database integrity verification, unit logic testing, API endpoint audit, end-to-end customer journey verification, owner kitchen kanban testing, defensive security testing, and data snapshot immutability verification. All 74 test cases have passed successfully with zero critical defects remaining.	RELEASE STATUS: VERIFIED & APPROVED Date: October 3, 2026 Healthy Bite Release v3.2
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